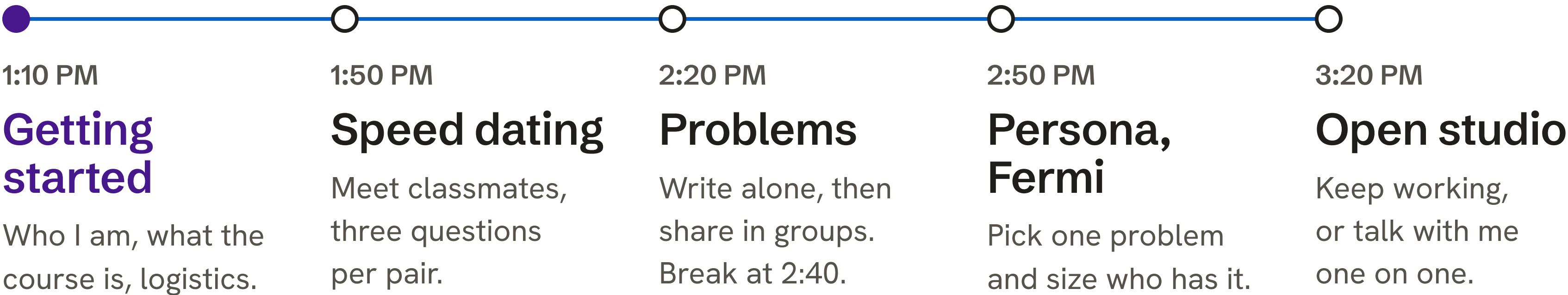


Startup Studio

AI-Accelerated Building and Validation

Class 1 · Friday, September 11, 2026 · 1:10 to 3:40 PM · Pupin 420

What we're doing today



Instructor Intro

Who am I?

Instructor Intro

Ken St. Clair

- Early-stage start-up Product Leader with 20+ years experience
 - Software engineering, sales engineering, product design, product leadership
 - Two robotics products from 0→1
 - Experience with 0→1, growth, acquisition, and deprecation
 - B2B, B2B2C, B2C
- BS Computer Science, Georgia Tech
- Masters of Human-Computer Interaction, Carnegie Mellon



aescape



Carnegie
Mellon
University



Georgia Institute
of Technology

Whom have I Worked with?

Instructor Intro

Select Partners and Clients

Gyms + Spas



Print Media



Broadcast



Streaming/Sports



Food Service

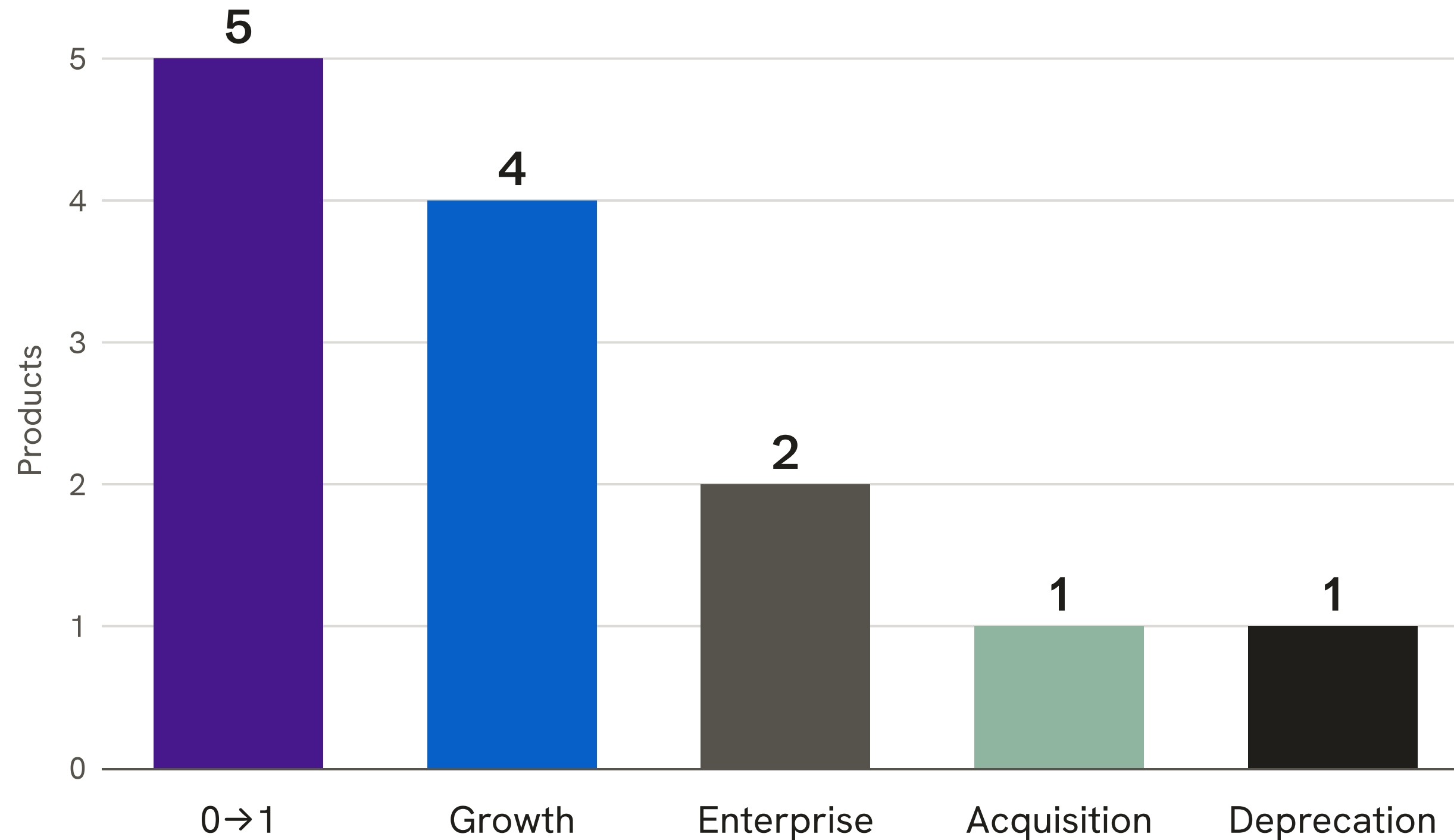


More...



Product Stages I've Worked On

Instructor Intro



Class Intro

What's a “Studio”

A course focused on **making**, **iterating**, and **critique**.

Students learn by building real artifacts and receiving continuous feedback, rather than by absorbing theory first.



What's “Startup Studio”

A course focused on **making, iterating, critique, experimentation and launching.**

Students learn by building real artifacts and receiving continuous feedback, rather than by absorbing theory first.



Three pillars of the course

Agentic engineering

Using AI tools to rapidly build software that is reliable and secure.

UX techniques

Methods for advancing product-market fit, including user interviews and usability testing.

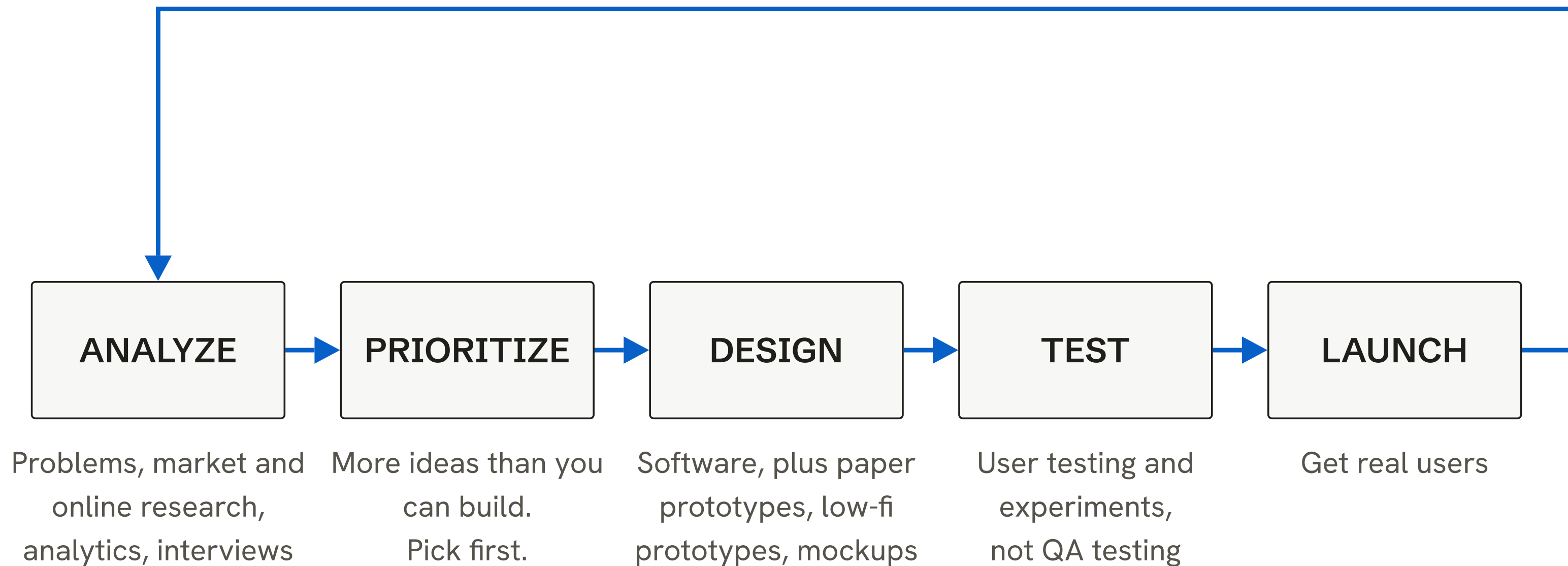
Startup knowledge

Growth hacking, presentation skills, fundraising, and how founders operate.

Startup OS

Class Intro

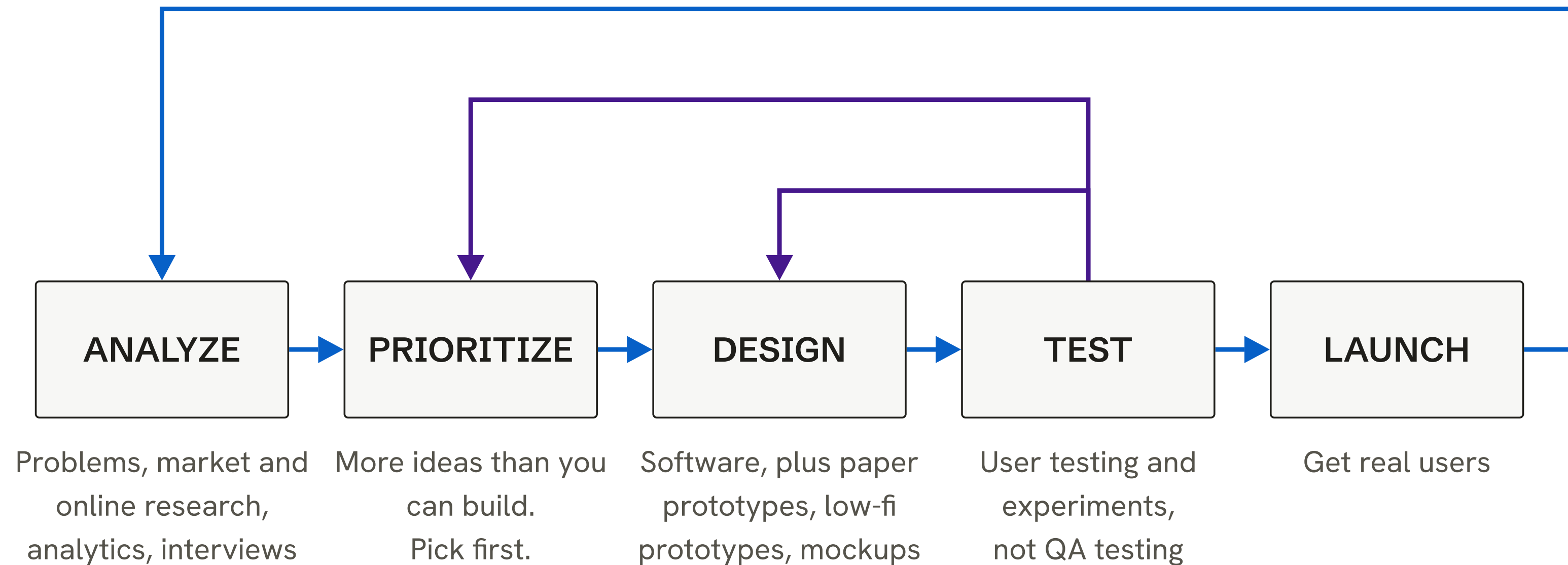
The operating system for this class



Startup OS

Class Intro

The operating system for this class



Analyze

Class Intro

- Problem Exploration
- Market research
- Online research
- Monitoring analytics
- User interviews

Prioritize

Class Intro

- I've got more ideas than I can execute at once
- How do I pick which to do first?

Design(/Build)

Class Intro

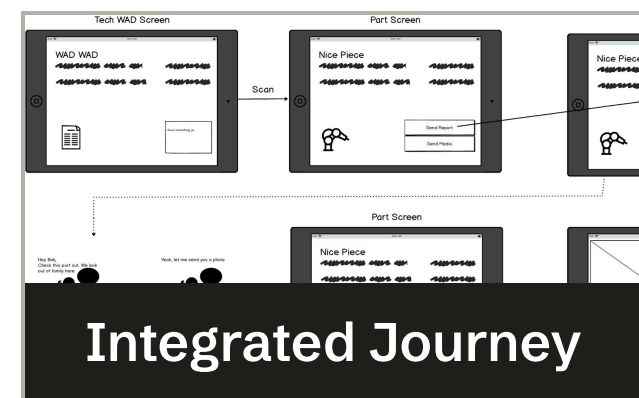
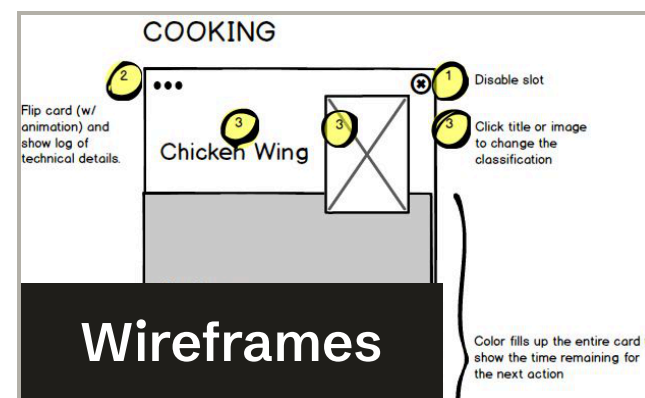
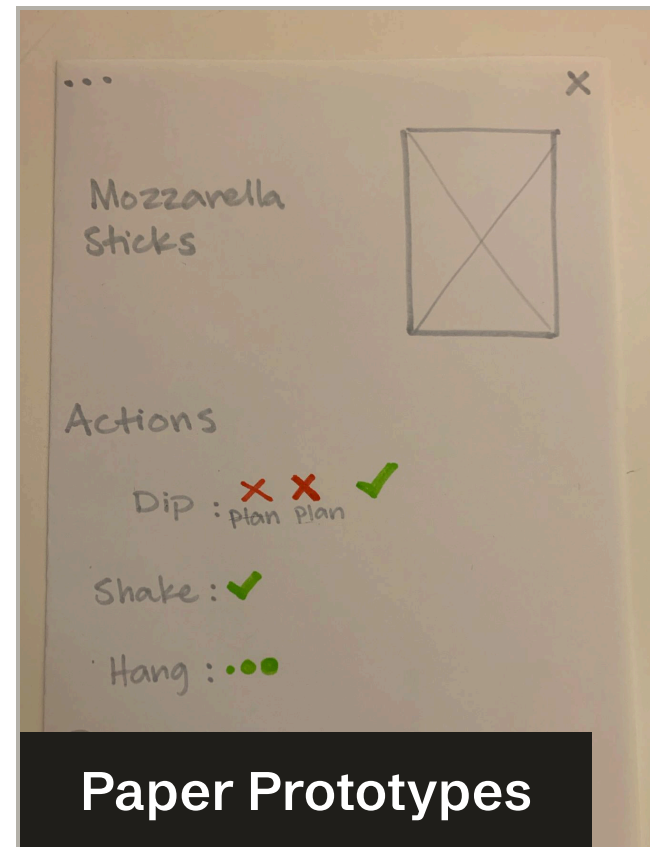
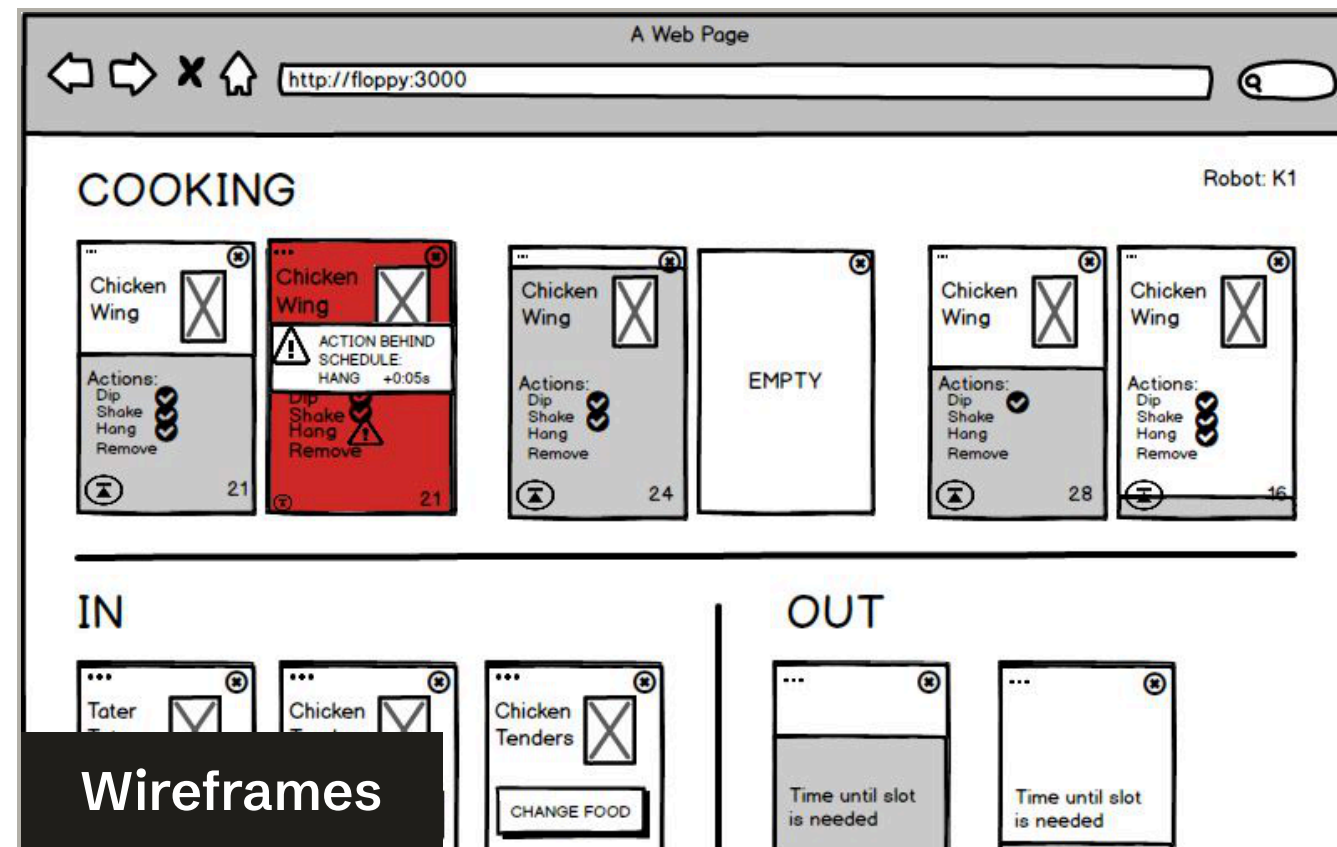
- Software, yes

But building also includes

- Paper prototypes
- Low-fi prototypes
- Mockups

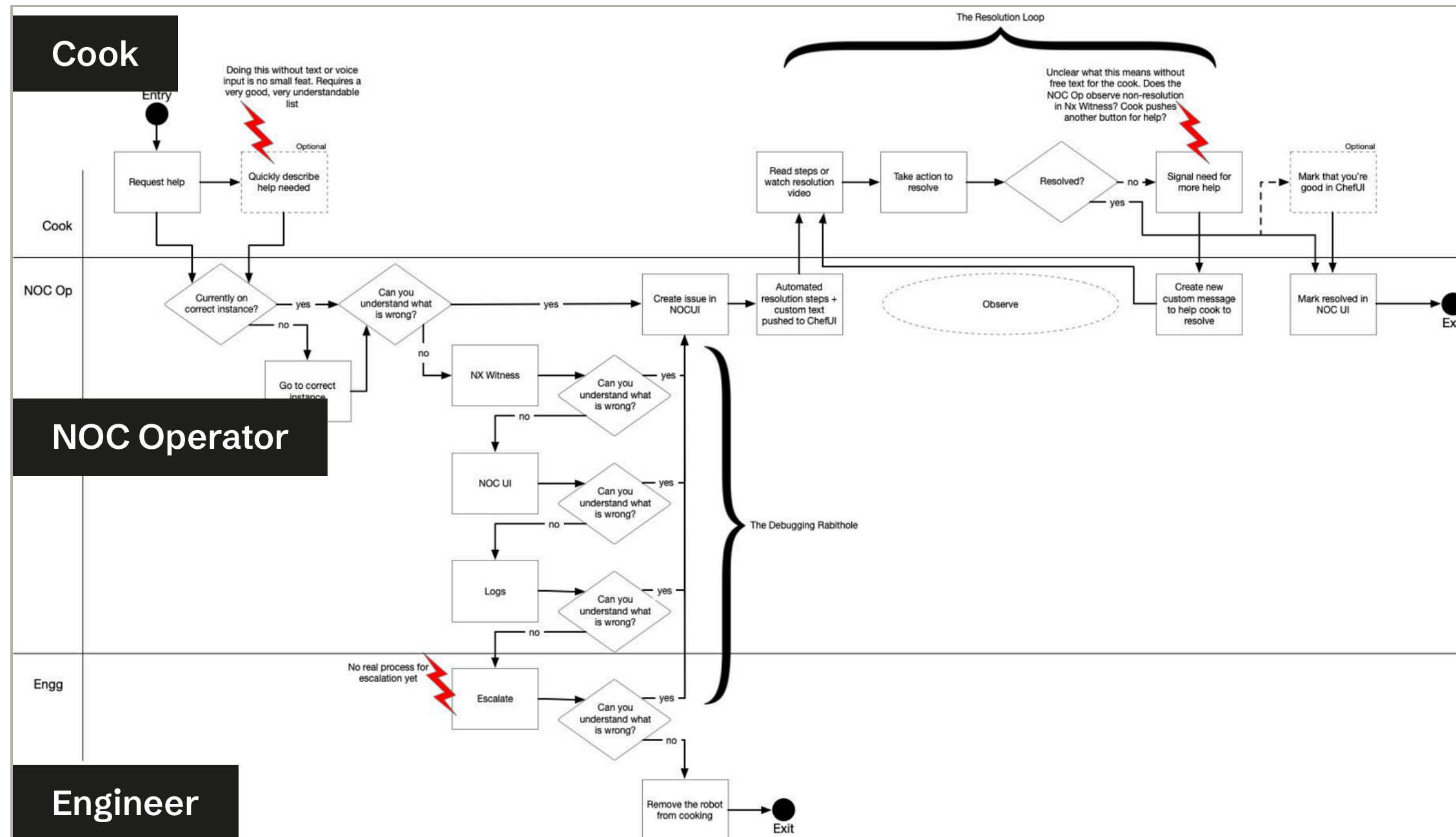
[Design Ex.] Personas, Wireframing, Integrated Journeys

Class Intro



[Design Ex.] The Debugging Rabbit Hole and Resolution Loop

Class Intro



Mapping the debugging process across each user – Cook, NOC Operator, and Engineer – allowed us to identify high impact areas of focus: the debugging rabbit hole, and the resolution communication loop.

Improved performance and efficiency in debug and resolution have a direct impact on revenue by affecting the throughput and overall volume of the cooking session.

Test

Class Intro

Not QA testing

- User Testing
- Experiments

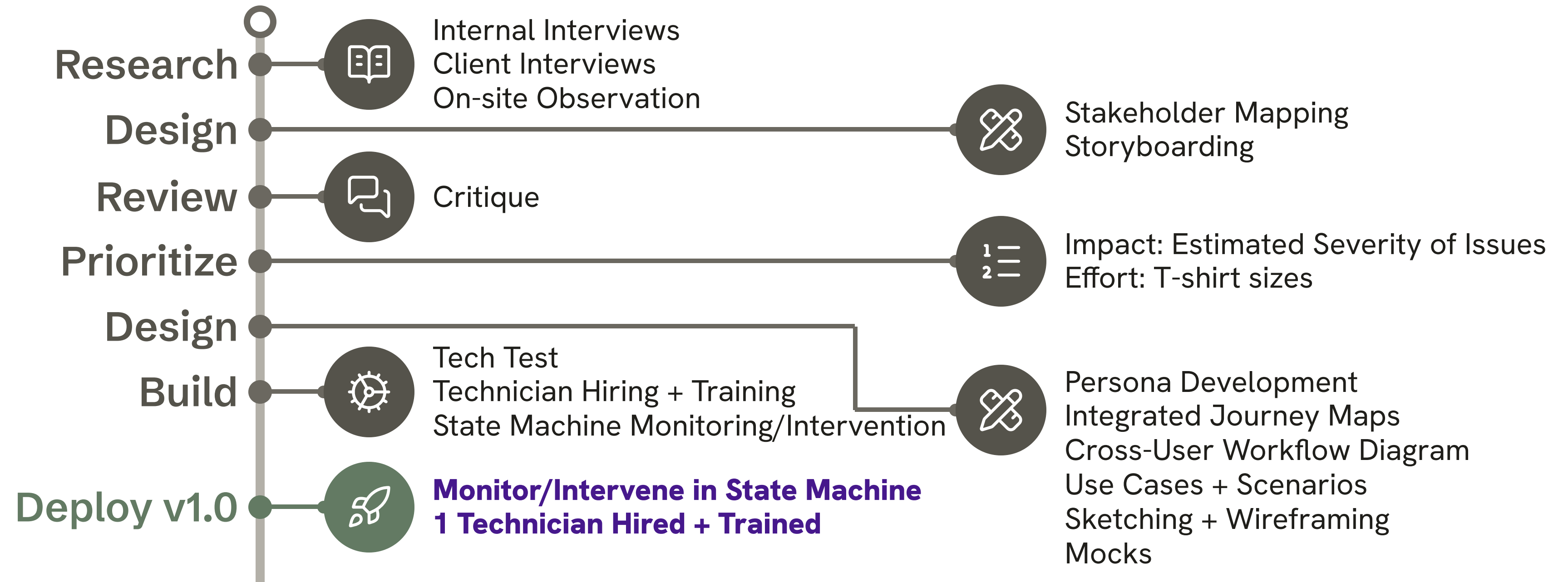
Launch

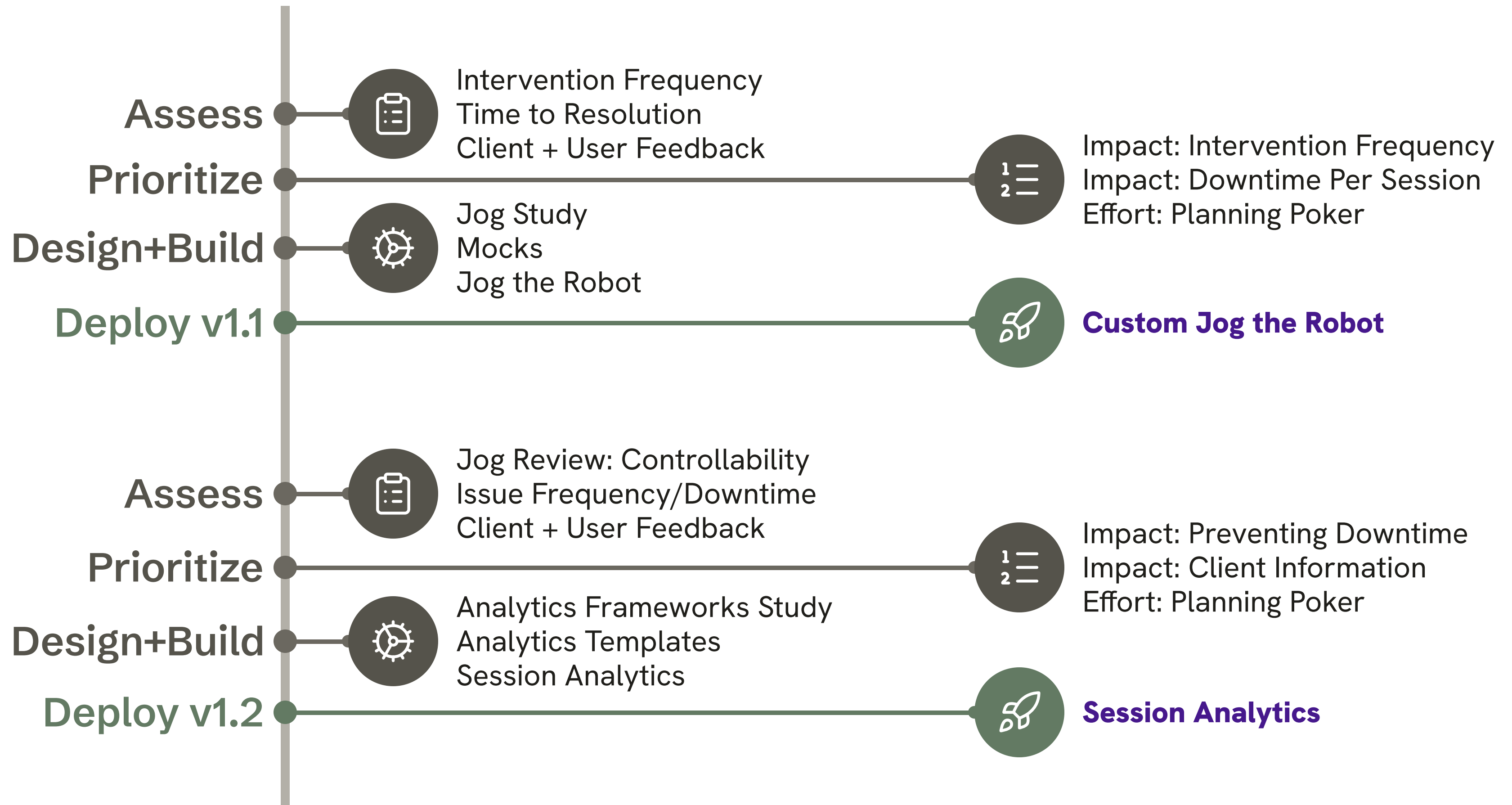
Class Intro

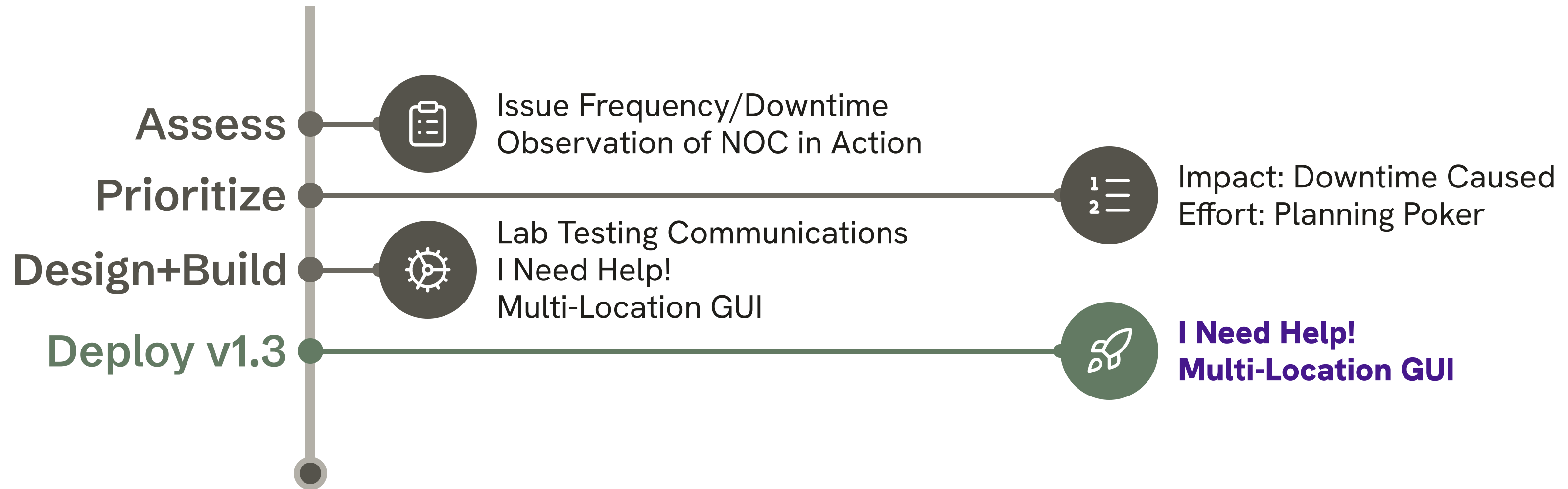
- Get real users

Real Timeline Excerpt: Network Operating Center

Class Intro







You will launch a real product with real users.

In teams, launched this semester,
pitched at Demo Day to guests.

What we build

- Products real users can sign up for this semester
- Web first. A native iOS app must launch on the App Store.
- B2B is fine if your team can get users onto it
- Bring your own idea or project. Talk to me about ownership.

What We're NOT Focusing On

- × Hardware
- × Medical diagnosis or treatment
- × Financial or legal advice
- × Anything with minors' data
- × Research and publishing
- × Deep theory
- B2B, kinda: only if your team can get users onto it

One 150-minute block every Friday

Class Intro

News segment

What changed this week in AI and startups.

Discussion

Talk the idea through together as a class.

Short talk

A new method or idea, kept brief.

Presentations

Teams show their work and get critique from the room.

Studio time

Try the method, build with your team, or meet with me.



This is not an **A+** course.

Your grade comes from Demo Day,
milestones, peer reviews, and real users.

Where your grade comes from

Class Intro

Component	What counts	Weight
Demo Day	Your pitch and working product	30%
Milestone submissions	On time, quality, shows learning	40%
Peer reviews	Your teammates rate your work	20%
User growth	Real users on your product	10%
Total		100%

Teamwork matters, and your team grades you

- Peer reviews are 20% of your grade.

5 Dysfunctions of a Team

- 1 Create Trust
- 2 Space for Healthy Conflict
- 3 Create Commitment
- 4 Peer Accountability
- 5 Attention to Results

What it costs

Item	Cost
Books	None. There are no books to buy.
A top AI coding tool subscription	Enough for real development work
Domain name	Under \$30 a year
Hosting	About \$5 a month
Team ad budget	\$100 per team, \$25 each
Interview incentives	Optional, up to \$100 per team

Working in Public

Class Intro

- Everything you build here is not stealth and may be shared publicly.
- Previous IP remains yours, but you should document it.
- Columbia doesn't claim what you build. This isn't an incubator.
- Ownership: see next slide.

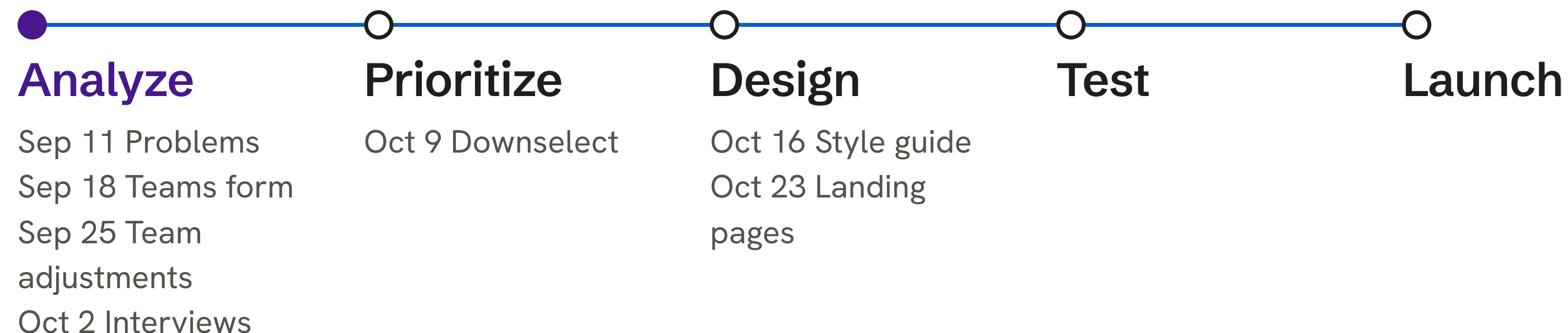
Ownership

Class Intro

- Company ownership usually comes as stock with a one-year cliff. Less than a year, and you get none of it.
- This course is about three months, so assume no one owns it.
- At the end, whoever wants to take it can.
- If you think you'll want it, talk about it early. The code base is easy to split. The user base is tougher.

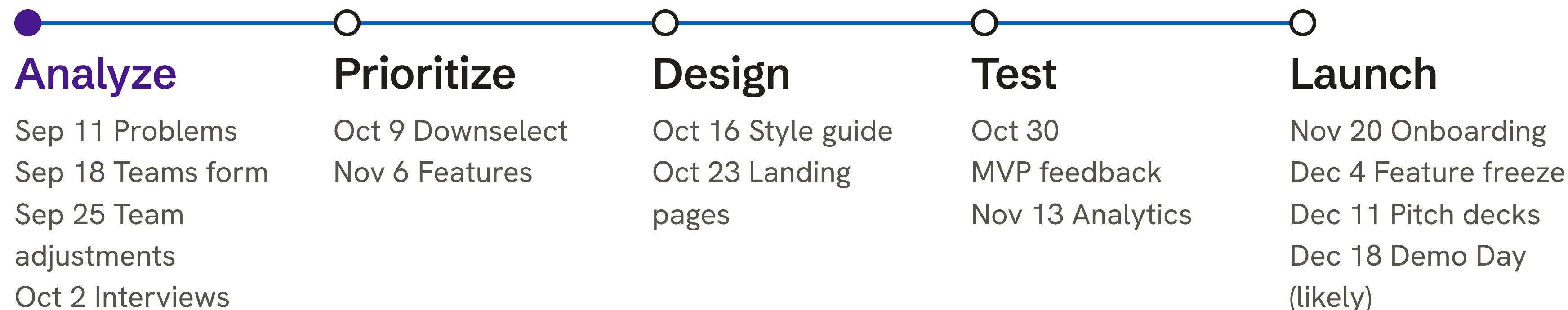
The semester, from problems to Demo Day

This schedule is a draft. Dates may change.



The semester, from problems to Demo Day

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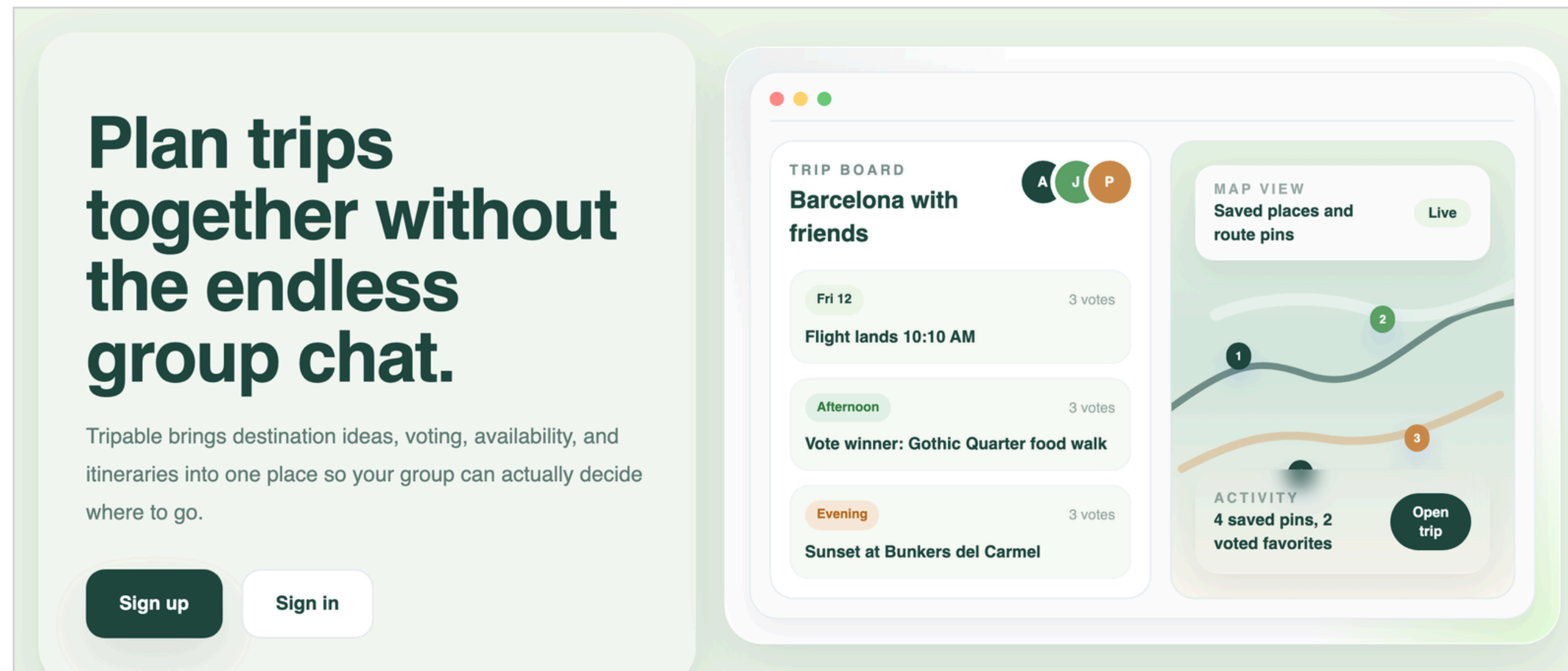


What students built last semester

Tripable

What students built last semester

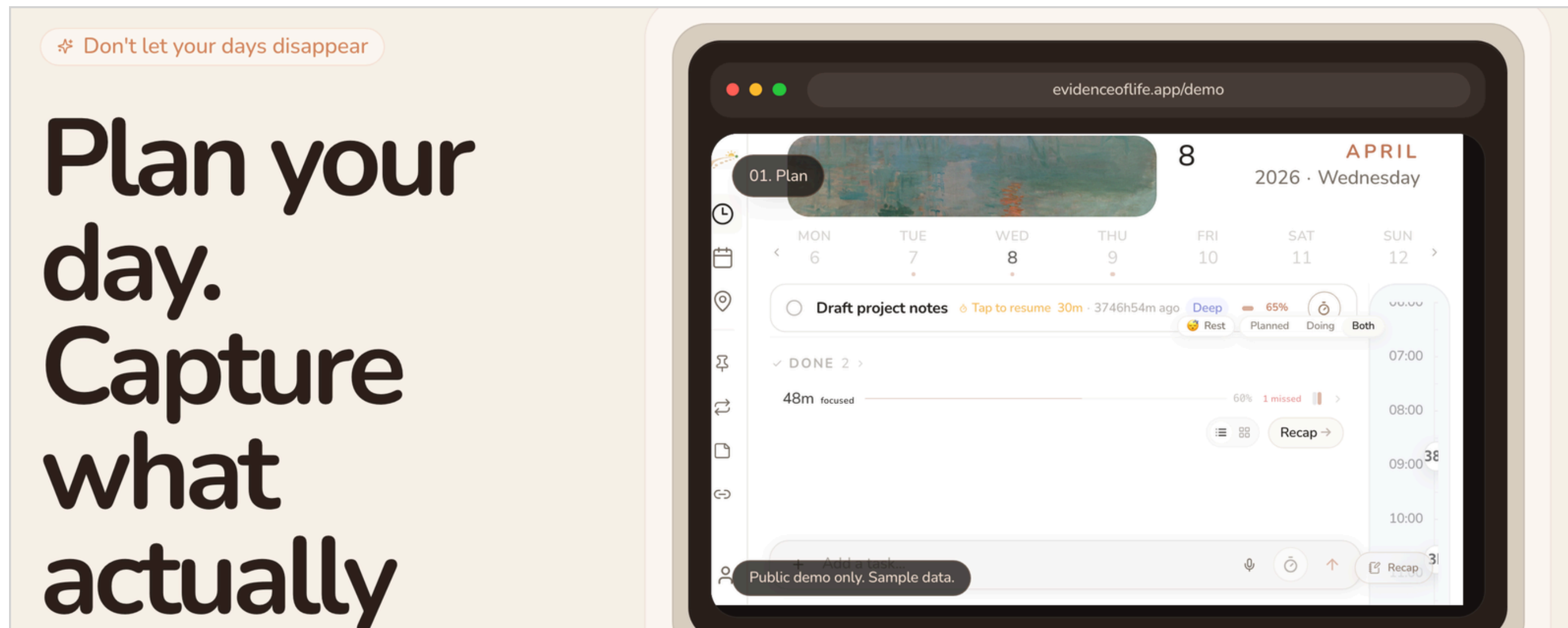
Group trip planning: votes, itinerary, and costs in one place.



Evidence of Life

What students built last semester

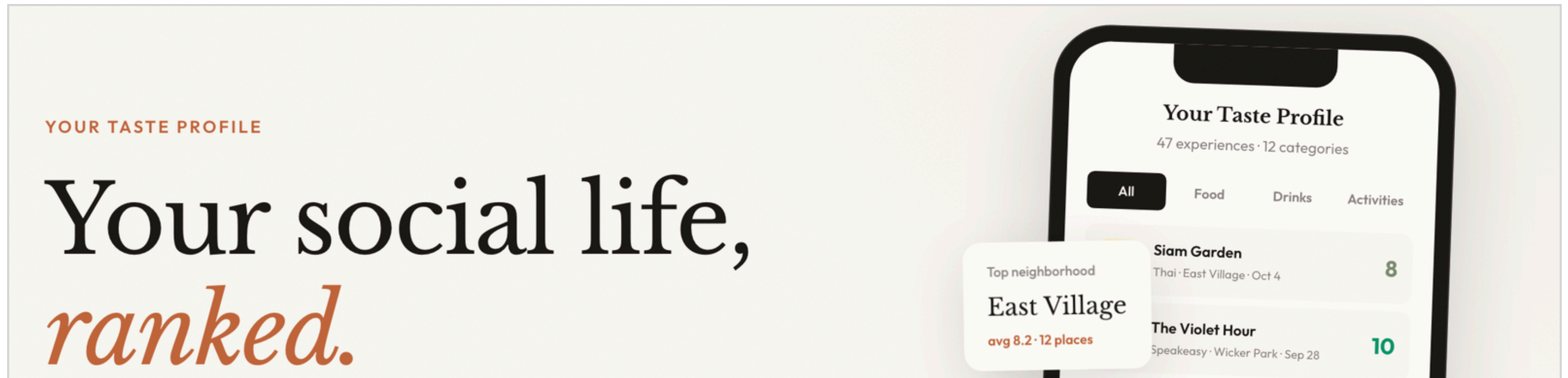
A private timeline that shows what actually happened,
not just what was planned.



Vouch

What students built last semester

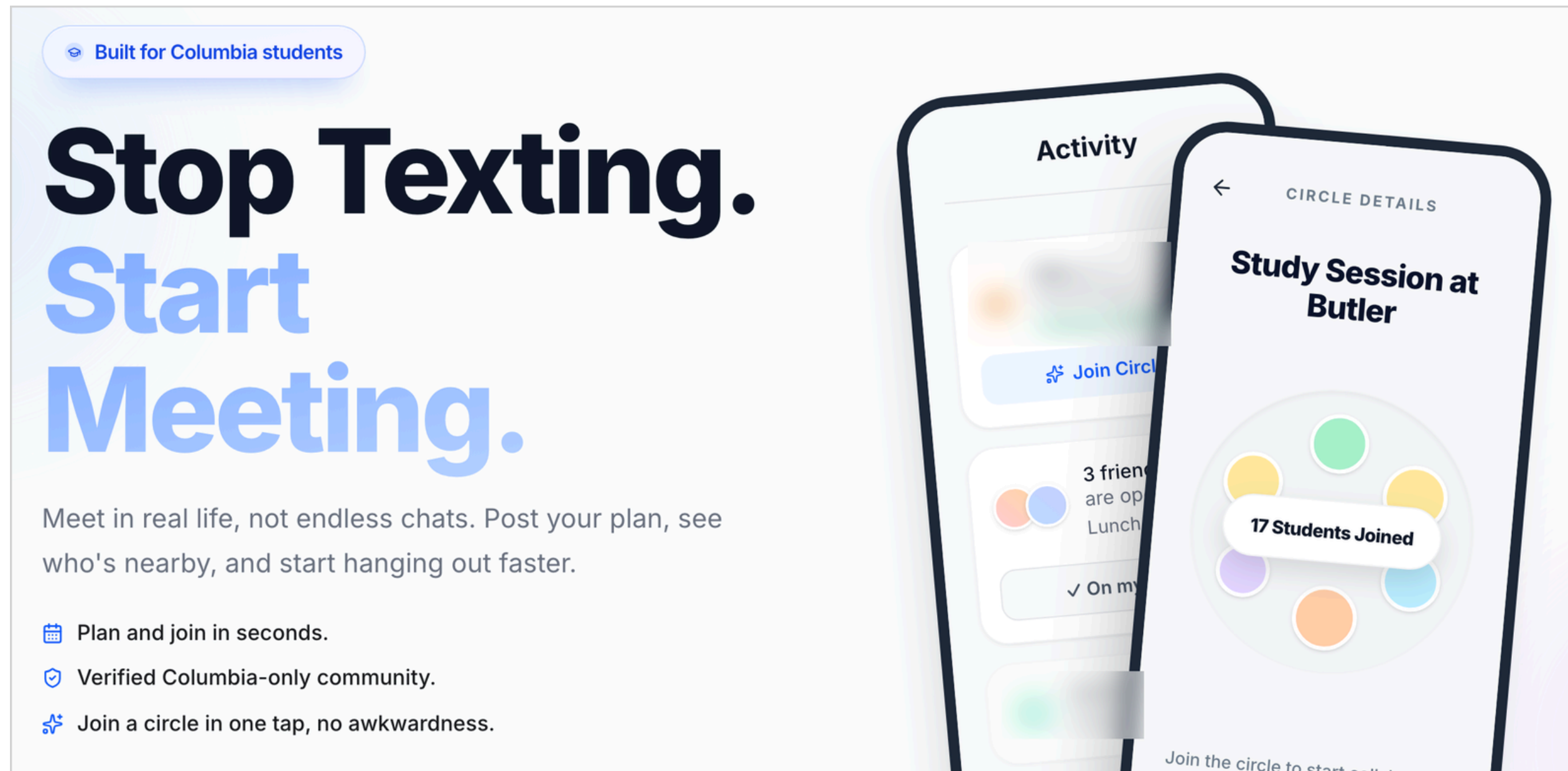
Ratings where your friends are the algorithm.



Circles

What students built last semester

Social coordination built for Columbia students.



QuestCity and Misfit.AI

What students built last semester

Stop Repeating the Same Date Night. Start Collecting Experiences.

Turn your city into an adventure game. Complete quests, discover hidden gems, and never run out of things to do—solo or together.

Get Early Access

3 Steps to More Adventures

Step 1: Pick a Quest

"Find 5 rooftop bars in Brooklyn" • "Hidden murals walking tour" • "Best tacos in a 2-mile radius"

QuestCity

Gamified outdoor exploration for friends and strangers.

One less decision every morning.

Wardrobe Planner looks at your calendar, checks the weather, and suggests exactly what to wear — using what you already own, so you show up looking effortless.

Connect Your Google Calendar — Start Free

No credit card required • \$1.99/mo after your free start • Cancel anytime

Misfit.AI

AI analyzes your wardrobe, calendar, and weather for wardrobe recommendations.

Logistics

Coming to class

Logistics

- Class is in person in Pupin 420. No remote attendance.
- No auditing.
- On the waitlist? Come today and on Sept 18.
- Sept 18 repeats today for anyone who joins late.

Class Position on AI

Logistics

Use AI for everything.

The AI you use will eventually come with a monetary cost.

Where things live

Logistics

Where	What's there
Courseworks	Syllabus, announcements, homework. COMSW4995_014_2026_3
Public repo	github.com/kenxle/columbia-startup-studio-fall-2026
Not added yet	Email me at krs2212@columbia.edu

Dec 18

Likely date for Demo Day

Likely Fri Dec 18, 12:00 to 3:00 PM, in the final exam slot, to be confirmed.

Nearest slot on the registrar's projected schedule

Studio time

People first, then problems, then one problem sized.

Speed dating: answer three questions

Studio time

- 1 What is one project you're proud of?
- 2 What kind of user or problem space are you drawn to?
- 3 What would you like to get better at through this course?

Speed dating

Studio time

30:00

- 1 What is one project you're proud of?
- 2 What kind of user or problem space are you drawn to?
- 3 What would you like to get better at through this course?
- 4 Switch partners when I call time, about every 7 minutes.

What makes a good problem here

Studio time

- ✓ A real person you could talk to this week
- ✓ A pain they already try to solve
- ✓ Something you could get users for in weeks
- ✗ Medical, financial, legal, or minors' data

Write problems, not solutions.

Problems brainstorm

Studio time

20:00

- 1 Alone and silent: write 3 problems.
5 minutes.
- 2 Form groups of 4 or 5. Share your problems.
- 3 Pick your group's top 2 or 3. 12 minutes.
- 4 Groups share. I capture the problems on screen and group them with the room.
3 minutes.

Back at 2:50.

Studio time

Look over the problems we collected.

Who has this problem?

Studio time

Pick one problem from the problems we collected.
Write a persona.

#	Field	What to capture
1	Who they are	A name, and what their day looks like
2	When it hits	The moment the problem shows up
3	What they tried	What they already do about it, and why it falls short

Fermi estimation: size the audience

Studio time

Break one big number into small guesses you can check.

#	Step	Estimate
1	Start big	US adults: 260M
2	Filter	× 40% live in cities = 104M
3	Narrow	× 15% order takeout weekly = 15.6M
4	Keep going	Add filters until you reach your persona

Check the pieces

Studio time

- 1 Pick one or two numbers in your chain.
- 2 Look up a real figure for each with a web search.
- 3 Do the multiplication yourselves.

Check the pieces, not the whole estimate.

Persona and Fermi sizing

Studio time

22:00

- 1 Pick one problem from the problems we collected.
- 2 Write the persona. 7 minutes.
- 3 Build a Fermi chain, then check the pieces with a web search. 12 minutes.
- 4 Circle your weakest assumption. 3 minutes.

Share your chain

Studio time

Two or three groups, 60 seconds each.

- Your persona in one sentence
- Your final number
- The assumption you trust least

Homework and open studio

Write in your voice

Homework and open studio

Everyone can smell AI slop now. Let's stop making it.

- Getting rid of AI tells is not as easy as it sounds.
- Figure out your own method. As sophisticated and multi-step as you like.
- **Few-shot examples:** give the AI real examples of your own writing.
- **Style guides**, banned words and phrases.

Come to class ready to talk about your method. We're comparing them.

Reading

Homework and open studio

Watch the official summary of The Five Dysfunctions of a Team.

youtube.com/watch?v=75bO_XWk7fw

The book is worth reading, but supplemental, not required.

This gives us a shared language for teamwork. Teams that work well together achieve more than individuals alone.

When it's due and where it goes

Homework and open studio

Due

Friday, Sept 18

- ✓ **Your setup:** the files that make the AI write like you. Skill files, instructions, prompt files, or the writing samples you gave it.
- ✓ **One output:** a LinkedIn post or article about AI, written by the AI in your voice, not over 500 words. A news update, an exploration of an AI topic, a position piece, or whatever you like.
- ✓ **On the waitlist?** You do it too. If you're not in Courseworks yet, email it to krs2212@columbia.edu.

Open studio: pick one

Homework and open studio



Finish chain

Look up real numbers for your Fermi chain with a web search.



Start homework

Pick your writing samples and start a prompt.



Find matches

Talk with someone from speed dating who shares your problem space.



Talk to me

Put your name and a short topic on the board.

See you **next Friday**

Sept 18 is the last day to add from the waitlist.

Class 2 · Friday, September 18 · 1:10 PM · Pupin 420